Forecasting Adjustment Mechanism Prices in France Using Econometric, Machine Learning and Deep Learning Models

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Contents

1 Introduction

French electricity market is a complex and dynamic system that plays a crucial role. Several modifications have occurred since 1999:

- 1999: Consumers of more than 100 GWh per year can choose their electricity suppliers. It is the first step of the french electricity market.
- 2000: The 100 GWh per year threshold has been decreased to 16 GWh per year
- 2004: All firms and collectivizes can chose their suppliers.
- 2007: The market is fully open.

Market is segmented into various markets based on the timing of electricity delivery, each with its unique characteristics and mechanisms. These segments include:

- Futures market. Allows one to buy or sell a certain amount of electricity (MWh) in advance. It is possible to buy a year in advance, a semester in advance, a month, a week, or a day. For example buying a CAL 27 means buying a power of 1 MW on all 2027. In terms of energy (MWh) this is equal to 1*365*24=8760 MWh. This market runs as an order book. The market is running by EEX (European Energy Exchange).
- Spot market. Allows to buy or sell for the next day. Runs as a blind auction. Bid and ask are sent to the EPEX SPOT market marker before 12 am on the previous day.
- Intraday markets. Allows to balance the position of suppliers to reduce the difference between position and real delivery. Runs as order books, but there are also auctions.
- Ancillary services. Run by the Transmission System Operator (TSO). Since electricity can not be enough stored, consumption must be equal to production. The key role of the TSO is to maintain this balance. For managing to keep this balance, TSO is able to pay one which is able to raise or decrease their production or consumption.
- Balancing markets. Managed by TSO. Balancing Responsible Party (BRP) must ensure that their positions matched real electricity consumption and production. The difference between both is paid for by the BRP. In France TSO is called "Reseau de Transport Electricté" RTE.

The Market chronology is summarized in the picture just below. The picture is extracted from "Modeling and optimal strategies in short-term energy markets" by Laura Tinsi.

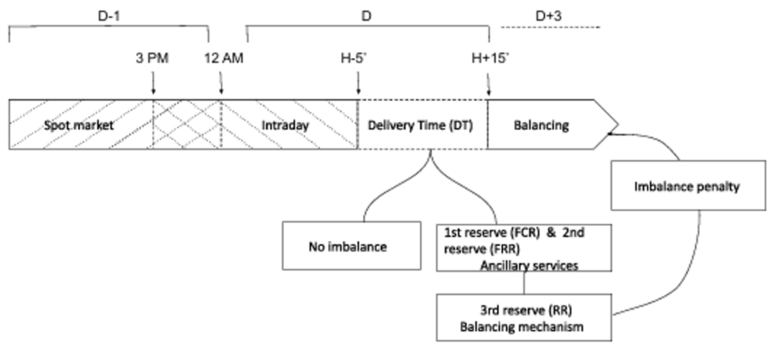

Figure 1: Market chronology

Recent literature on electricity markets highlights a diversity of approaches for price forecasting, ranging from traditional econometric models to advanced machine learning (ML) and deep learning (DL) techniques. For the spot market, studies such as those by Vassiloupoulos and Dutot et al. emphasize the importance of structural models, equilibrium dynamics, and hybrid approaches combining econometric models with ML techniques. Weron's synthesis of model evolution underscores the potential of hybrid approaches for improved forecasting accuracy. For the ancillary services, research by Deng et al. demonstrates the superiority of deep learning models, such as LSTM, in capturing complex seasonality and non-nonlinearities, outperforming traditional methods.

This thesis aims to answer the research question: "Can ancillary markets forecasting be used to maximize profits?" To address this question, several sub-questions will be explored:

- Which models can forecast ancillary market prices, and for which time horizon?
- Is the forecasting accuracy sufficient to generate profits?
- Is the forecasting enough tp generate more profits than with a naive model?

In this paper several parameters are not taken into account:

- Market fees are not considered,
- It is assumed that the TSO merit order is only based on prices.

The methodology for this study involves several steps:

- Collecting price data,
- Prepossessing and visualizing the data,

- Collecting production and consumption data,
- Testing and evaluating deep learning, econometric, and machine learning models,
- Selecting the best model and conducting back testing to assess its effectiveness in the sample,
- Selecting the best model and conducting back testing to assess its effectiveness out of the sample.

This thesis will contribute to the existing body of knowledge by providing insight into the application of forecasting models in the ancillary electricity markets to maximize profits. By evaluating various models and their accuracy, this research aims to offer practical recommendations for market participants and operators of energy systems.

2 Literature Review

2.1 Introduction

One main point to emphasize is that the way to be paid is different in spot and ancillary market. Indeed in spot market, one is payed according to marginal price. Meaning, the need of prices forecasting is not obvious for a producer since it is produced only if it is profitable and if the marginal price is above the price asked producer payed at this price. On ancillary markets, it is different since it is payed the price asked. The need of forecasting is tremendous for raising the profitability.
Moreover the inherent volatility of the electricity market, amplified by the intermittency of renewable energies and recent exogenous shocks (e.g., COVID-19 pandemic, war in Ukraine), has imposed new modeling needs for electricity prices. Faced with this increasing complexity, traditional forecasting methods show their limits, both in terms of accuracy and robustness to regime changes. In this context, two contemporary approaches stand out: on the one hand, the integration of seasonal attention mechanisms in deep learning models to improve balancing price forecasts in the UK (Deng et al., 2024), and on the other hand, adaptive probabilistic forecasting applied to the French market, combining online aggregation and conformal prediction (Dutot et al., 2024).

2.2 Balancing Price Forecasting with Seasonal Attention (SA-BiLSTM)

In article published in *Energy Economics*, DENG (2024) focuses on forecasting balancing prices in the UK market, which plays a crucial role in the stability of the electrical grid. The authors propose a BiLSTM (Bidirectional Long Short-Term Memory) architecture enriched with a seasonal attention mechanism, the SA-BiLSTM, to capture daily recurrences (e.g., the 48 half-hours of the British system).

The modeling is based on a rich set of explanatory variables extracted from the BMRS platform from 2016 to 2019: demand forecast errors, wind and solar forecast errors, reserve margin, exchange volumes, etc. This framework is relevant because it explicitly takes into account the temporal structure and seasonal effects inherent in electricity prices.

The empirical results show that the SA-BiLSTM model significantly outperforms ten other state-of-the-art approaches (ANN, GRU, LSTM, TCN, Transformer, SVR, XGBoost, etc.) in terms of RMSE, MAE, and MAPE, particularly in forecasting price peaks. Seasonal attention enhances the model's sensitivity to intra-daily cycles and improves out-of-sample stability with a predictability gain of 11% to 15% compared to benchmarks.

2.3 Adaptive Probabilistic Forecasting of French Spot Prices

DUTOT (2024) adopts a probabilistic forecasting approach applied to the French spot market (EPEX). The authors use past prices, commodity prices, cross-border exchanges, and a new variable with high explanatory value: nuclear availability. The study focuses on the years 2020–2021, marked by extreme turbulence (COVID, nuclear corrosion, gas crisis).

Several quantile regression methods are used (QRF, QGAM, QGB, Lasso QR, Linear QR). A conformal prediction layer is also added, making the predictions probabilistic. These models are then dynamically adjusted.

This combination leads to a significant improvement in the validity and precision of probabilistic intervals, especially after 2021. The approach maintains coverage close to the nominal (90–95%) while keeping a reasonable interval width.

In this thesis, it is proposed to test the various models mentioned on the French adjustment mechanism and evaluate the interest of probabilistic models that adjust dynamically.

2.4 Methodological Comparison

The two contributions target similar horizons (short-term) and rely on advanced machine learning techniques. However, they differ fundamentally in their objectives and methodologies.

- **Type of Output**: The SA-BiLSTM model aims for point forecasts, while the French approaches produce probabilistic forecasts in the form of intervals.

- **Temporal Stability**: SA-BiLSTM is trained on a fixed history without dynamic recalibration. In contrast, Dutot et al.'s strategy relies on continuous adjustments (rolling window, OSSCP).

- **Shock Management**: The French approach is explicitly designed to handle regime shifts through adaptive correctives. SA-BiLSTM, on the other hand, remains performant on stationary data but may be limited during severe exogenous disruptions.

- **Computational Complexity**: SA-BiLSTM is costly to train but fast in inference. Adaptive methods require constant recalibration, with a trade-off between reactivity and complexity.

2.5 Limitations and Perspectives

Despite their notable contributions, both studies have certain limitations. On the SA-BiLSTM side, the absence of probabilistic forecasting limits its use in risk management. Extending the model to longer horizons or other markets (e.g., France, Germany) remains to be explored.

For the probabilistic approach, performance heavily depends on the quality of the base models (experts). Additionally, implementing OSSCP-horizon requires fine calibration of learning and validation windows, as well as significant computational resources.

Interesting future directions include:

- Hybridizing the two paradigms: integrating an attention mechanism into an adaptive probabilistic forecasting framework.
- Exploiting real-time exogenous data (e.g., weather, nuclear unavailability) to enhance model adaptability.

2.6 In Summary

Recent literature on electricity price forecasting presents increasingly sophisticated approaches, combining deep learning, probabilistic prediction, and dynamic adaptability. This thesis aims to continue the two analyzed contributions. On the one hand, the SA-BiLSTM model shows that seasonal attention improves the accuracy of extreme forecasts. On the other hand, the integration of conformal prediction and online aggregation offers a robust and reactive solution to non-stationarity.

The future of energy forecasting models seems to lie in the convergence of these approaches, capable of capturing complex temporal dynamics, anticipating extremes, and continuously adapting to market changes.

The main objective of this document is to use the different models explained above on the French balancing market.

3 Data

3.1 Data Used for the Thesis

In this thesis, the data to be forecasted are related to the french ancillary services. One mechanism of these services is called "Mécanisme d'Ajustement" (meaning Adjustment Mechanism) in France. It helps maintaining the balance of the electrical grid. This mechanism helps to ensure that electricity production matches consumption in real-time, a crucial challenge given that electricity cannot be stored efficiently on a large scale. In France, this mechanism is managed by RTE, the Electricity Transmission Network.

3.2 RTE and its Role

RTE, or Réseau de Transport d'Électricité, is the operator of the electricity transmission network in France. It is responsible for managing and operating the high-voltage network, thus ensuring the balance between electricity supply and demand in real-time. An unbalance could result in a raise or decrease of the frequency grid which could create a black out. RTE plays a key role in the stability of the French electrical system, ensuring that electricity is transported from producers to consumers in a reliable and secure manner. Among its many responsibilities, RTE also manages the adjustment mechanism, a market where producers and consumers can adjust their positions up to 30 minutes before delivery. This mechanism operates on a pay-as-bid system, where participants submit bids to adjust production or consumption and are paid according to the prices they proposed.

3.3 Electricity Production Data

Electricity production data by type of energy are essential for understanding and forecasting prices in the adjustment market. The main types of energy in France include:

- **Solar**: Solar energy is produced from photovoltaic panels that convert sunlight into electricity. Solar production is intermittent and highly dependent on weather conditions.

- **Onshore Wind**: Onshore wind energy is produced by wind turbines located on land. These installations often benefit from more stable winds than those at sea but may be limited by space constraints and local regulations. Production is intermittent and highly dependent on weather conditions.

- **Offshore Wind**: Offshore wind energy is produced by wind turbines installed in coastal waters. These installations can capture stronger and more constant winds but face greater technical and logistical challenges. Production is intermittent and highly dependent on weather conditions.

- **Biomass**: Biomass uses organic materials, such as wood, agricultural waste, and food waste, to produce electricity. This energy source is more stable than solar or wind but its production can vary depending on resource availability.
- **Geothermal**: Geothermal energy is produced by exploiting the Earth's internal heat. It is generally stable and continuous but its availability depends on local geological characteristics.
- **Nuclear**: Nuclear energy is produced in nuclear power plants where energy is generated by the fission of uranium atoms. In France, nuclear power accounts for a significant portion of total electricity production and is a stable and continuous energy source.
- **Coal**: Although less used in France, coal remains an important fossil energy source in some countries. It is burned to produce steam that drives electricity-generating turbines.
- **Natural Gas**: Natural gas is a fossil energy source used to produce electricity in combined cycle or gas turbine power plants. It is cleaner than coal but remains a fossil fuel.
- **Oil**: Oil is another fossil energy source, mainly used in thermal power plants to produce electricity. Its use is declining due to its environmental impact and cost.

3.4 Electricity Consumption Data

Electricity consumption is another key element for price forecasting in the adjustment market. Electricity consumption varies depending on several factors, including:

- **Temperature**: Electricity consumption is strongly influenced by weather conditions. For example, periods of intense cold in winter or extreme heat in summer can lead to increased consumption due to heating or air conditioning.
- **Economic Activity**: Levels of economic activity also influence electricity consumption. Weekdays, for example, generally have higher consumption than weekends.
- **Hours of the Day**: Consumption also varies depending on the time of day, with peaks generally observed in the morning and evening.

3.5 Data Retrieval

The data used in this thesis are retrieved via the APIs provided by RTE, following the RTE API user guides. These APIs allow direct and structured access to production and consumption data, facilitating the analysis and modeling of electricity market trends.

By using this production and consumption data, we aim to develop forecasting models that can help optimize decisions in the adjustment market, thereby contributing to the stability and efficiency of the French electrical grid.

3.6 Data Visualization

3.6.1 Distribution of the MA Variable

The main statistical data related to the MA price for the year 2024 are displayed below:

Statistic	**Value**
Mean	79.50
Standard Deviation	111.90
Minimum	-2917.98
1st Quartile (25%)	18.70
Median (50%)	67.65
3rd Quartile (75%)	115.90
Maximum	3866.00

Table 1: Descriptive statistics of the `MA` variable

The statistical analysis of the `MA` variable, summarized in Table 1, reveals an asymmetric distribution characterized by significant outliers. Indeed, although the median is relatively moderate (67.65), the maximum value reaches 3866, while the minimum drops to -2917.98. This marked contrast between the extremes and the quartiles (25% = 18.70; 75% = 115.90) indicates a high dispersion of the data. The high standard deviation (111.90) relative to the mean (79.50) also confirms the presence of atypical observations likely to skew the analyses if not handled with caution. These results suggest the relatively frequent existence of extreme values.

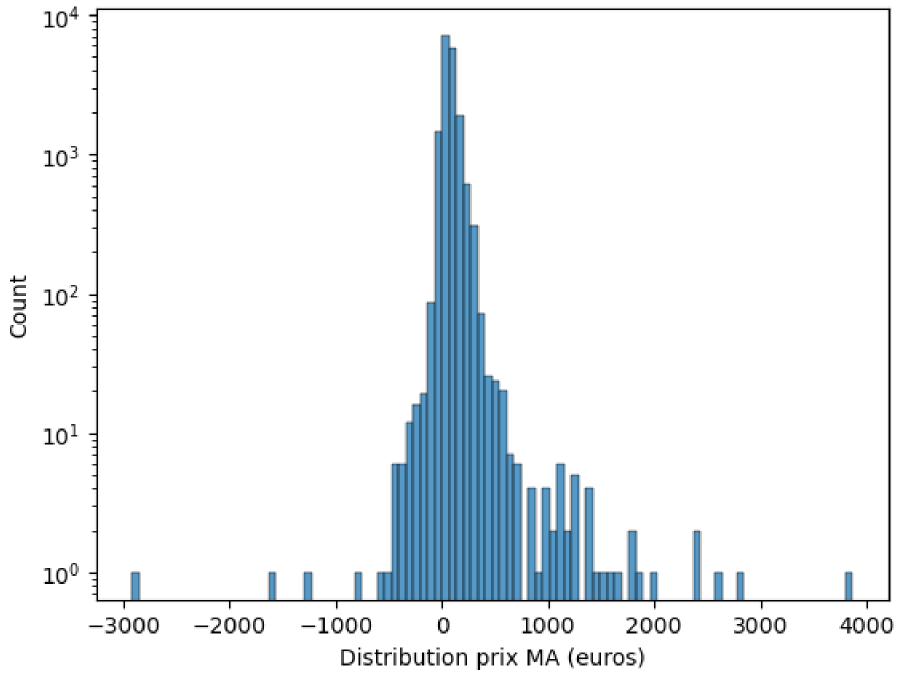

Figure 2: Distribution of MA Prices

The figure above confirms the presence of fat tails with prices centered around 80 euros.

3.6.2 Distribution of Production

In France, electricity is produced from three main energy sources: nuclear, renewable, and fossil. Nuclear energy, which accounts for a large part of French electricity production, comes from the fission of uranium atoms in nuclear power plants. This method is very efficient and produces a large amount of energy without direct greenhouse gas emissions, but it poses challenges in terms of radioactive waste management and nuclear accident risks.

Renewable energies include sources such as hydroelectric, wind, solar, and biomass. These sources are inexhaustible and have a lower environmental impact compared to other types of energy, as they emit very little greenhouse gas. However, their production can be intermittent, depending on weather conditions, and their development often requires significant initial investments.

Finally, fossil energies, such as coal, oil, and natural gas, are used to produce electricity in thermal power plants. These sources are abundant and relatively inexpensive to exploit, but their combustion releases significant amounts of carbon dioxide, contributing to global warming. Additionally, fossil resources are limited, and their extraction can have negative environmental and social impacts.

Each of these energy sources has its advantages and disadvantages, and the French energy mix aims to balance these different options to ensure stable, sustainable, and environmentally friendly electricity production.

The graph below shows electricity production in France in 2024:

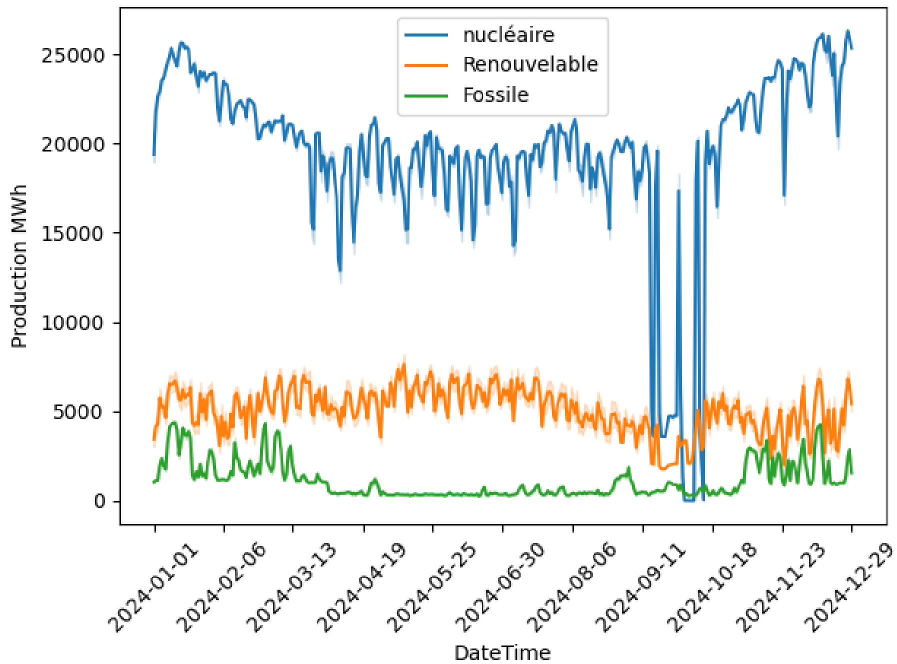

Figure 3: Electricity Production in France in 2024

As shown in the graph above, the French electrical system relies mainly on nuclear energy to ensure stable and continuous electricity production. This energy source forms the backbone of the national electrical grid, covering a large part of the country's energy needs reliably and with relatively low greenhouse gas emissions. Renewable energies, although intermittent, play a crucial complementary role. Their use is optimized whenever they are available, thanks to their increasingly competitive production costs and low environmental impact.

During periods of high demand, particularly during winter cold spells, fossil energies are called upon to provide the necessary additional supply. Although their use is limited due to their higher environmental impact, they offer essential flexibility to meet consumption peaks and ensure the stability of the electrical grid. Thus, the French energy mix, as illustrated by the graph, effectively combines these different sources to ensure a balanced, sustainable, and secure electricity supply.

3.6.3 Distribution of Consumption

The reason for the production distribution seen above is consumption. Indeed, in winter, the cold implies a greater need for electricity to ensure lighting and heating. This need translates into higher consumption. In summer, consumption is much lower.

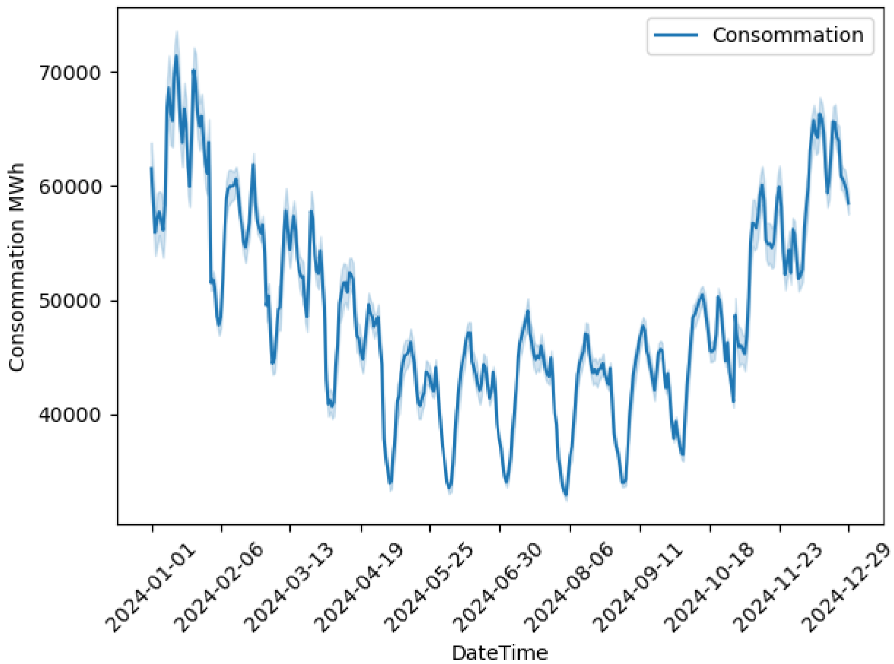

Figure 4: Electricity Consumption in 2024

3.6.4 Correlation

It is proposed to display the correlation between all the variables discussed in the paragraphs above as well as the spot price. The following heatmap is obtained:

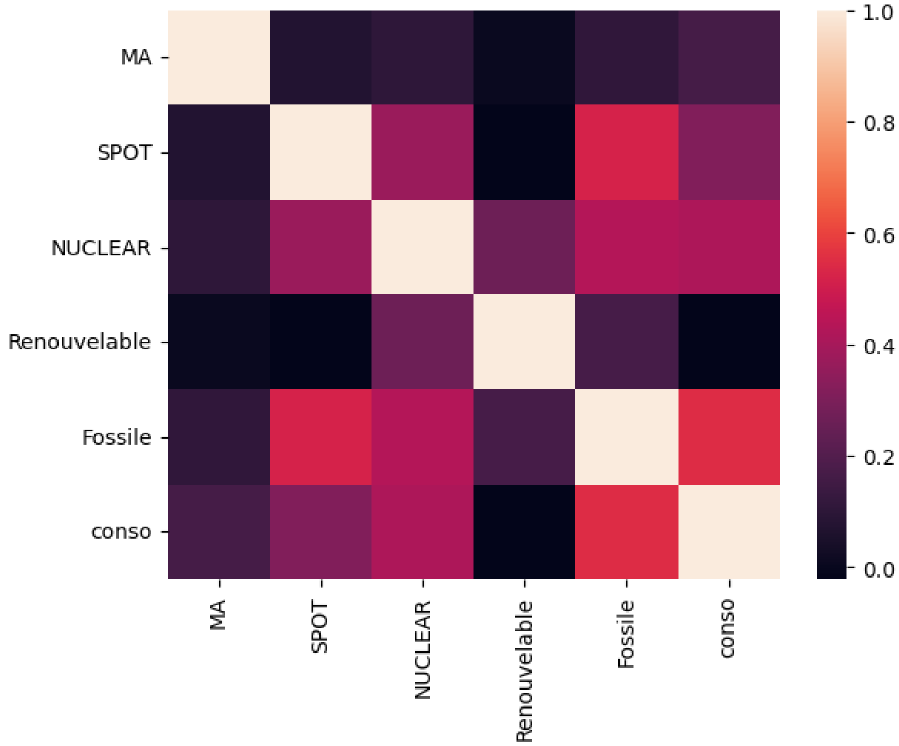

Figure 5: Heatmap of Correlations

It is observed that the price of the adjustment mechanism (MA) has a moderate correlation with other variables, suggesting that it is influenced by various market factors. For example, a positive correlation with SPOT prices indicates that price adjustments tend to follow trends in the wholesale market, reflecting real-time variations in supply and demand. This can be attributed to the fact that the adjustment mechanism is designed to respond to market imbalances, encouraging producers to adjust their production according to needs.

The correlation between the price of the adjustment mechanism and electricity production from nuclear and renewable sources seems to be weak, which may indicate that these production sources, often less costly and more stable, have a limited impact on price adjustments. On the other hand, a more marked correlation with fossil production could reflect the higher cost and flexibility of these energy sources, which are often used to meet peak demand.

Finally, the relationship between the adjustment mechanism and electricity consumption (conso) highlights the importance of this mechanism in managing demand fluctuations. By adjusting prices according to consumption, the adjustment mechanism helps ensure a stable and efficient electricity supply while reflecting the actual costs of production. To date, different models have been used to attempt to forecast MA prices. We will divide the model in three part:

- **Deep learning**
- **Machine learning**
- **Machine learning**

4 Deep learning

Deep learning consists in making an artificial neurons by implementing an activation function. Let us define the following:

- $\mathbf{x}$: the input vector
- $\mathbf{y}$: the output vector
- $\mathbf{w}$: the weight vector
- $\sigma(\mathbf{x})$: the activation function applied to $\mathbf{x}$

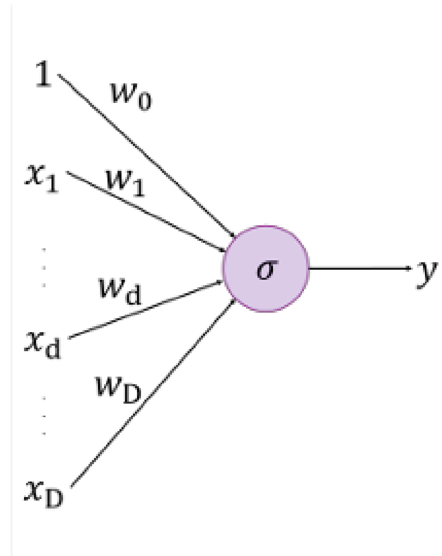

Figure 6: One neural

The objective is to minimize the difference between the output vector computed from the input vector and the true value of y. This minimization is achieved by appropriately selecting the weight vector $\mathbf{w}$. Between input vector and the output layer there are hidden layers that allow us to use different activation functions. The all is the neural network.

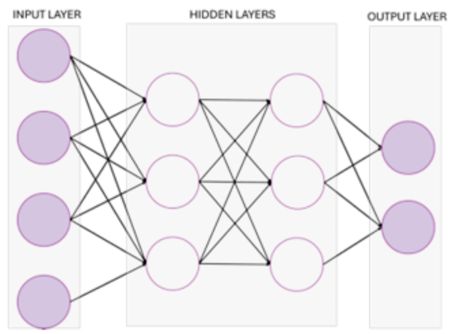

Figure 7: Neural network

4.1 Neural Networks

Neural networks can be segregated into two kinds. One which do not has memory. There is no cycle, information flow form input to output layers. These are called Feed-Forward networks. In theory they are not the best fitted for time series. The other kind is networks with cycles, previous information is used for current prediction. More suitable for time series. For this study we will use 5 neural network:

- **Feed Forward Network**
- **Recurrent Neural Network**
- **Long Short Term Memory**
- **Self-Attention Long Short Term Memory**
- **Bidirectional Self-Attention Long Short Term Memory**

4.2 Feed Forward Network

One of the simplest kind of neural network. If we consider the neural network in the figure 6, the equations resolved would be for each layer:

- $x^{(1)} = \sigma^{(1)} \left(\mathbf{W}^{(1)} \cdot \mathbf{x} + \mathbf{b}^{(1)} \right)$
- $x^{(2)} = \sigma^{(2)} \left(\mathbf{W}^{(2)} \cdot x^{(1)} + \mathbf{b}^{(2)} \right)$

where $\mathbf{W}$ is the weight vector for each layer, and b a constant.

4.3 Recurrent Neural Network

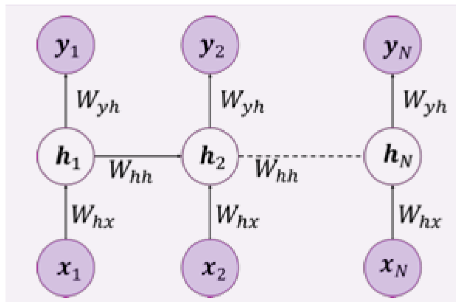

Figure 8: Recurrent Neural Network

RNN have been introduced by Elman in 1990. If we consider a single RNN cell unrolled over time, the equations resolved at each time step t are:

$$\mathbf{h}^{(t)} = \sigma\left(\mathbf{W}_x \cdot \mathbf{x}^{(t)} + \mathbf{W}_h \cdot \mathbf{h}^{(t-1)} + \mathbf{b}\right)$$

Only the final hidden state $\mathbf{h}^{(n)}$ is used to compute the output:

$$\mathbf{y} = \phi\left(\mathbf{W}_y \cdot \mathbf{h}^{(n)} + \mathbf{b}_y\right)$$

where:

- $\mathbf{x}^{(t)}$ is the input at time step t,
- $\mathbf{h}^{(t)}$ is the hidden state at time t,
- $\mathbf{W}_x$, $\mathbf{W}_h$, and $\mathbf{W}_y$ are the weight vectors or matrices,
- $\mathbf{b}$ and $\mathbf{b}_y$ are bias vectors,
- σ is the activation function for the hidden state (e.g., tanh or ReLU),
- ϕ is the activation function for the output (e.g., softmax or linear).

The main issue with this kind of model is the multiplication of gradient terms. The gradient ends up becoming infinitely small or large. Resulting in impossible training. The solution is Long Short Term Memory architecture.

4.4 Long Short Term Memory

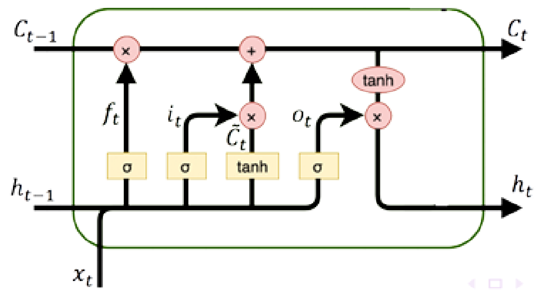

Figure 9: Long Short Term Memory

Long Short-Term Memory networks (LSTM) were introduced by Hochreiter and Schmidhuber (1997). The vector $\mathbf{C}_t$ represents the memory cell of the LSTM at time t. It stores information based on the current input $\mathbf{x}_t$ and the previous hidden state $\mathbf{h}_{t-1}$. The hidden state $\mathbf{h}_t$ at time t is computed from the cell state $\mathbf{C}_t$, and it carries the output information of the LSTM.

This information flow is carefully regulated by three gates:

- the input gate $\mathbf{i}_t$, which controls how much new information is added to the cell,
- the forget gate $\mathbf{f}_t$, which decides what part of the previous memory should be discarded,
- and the output gate $\mathbf{o}_t$, which determines how much of the cell state is passed to the hidden state.

The equations are:

- Input gate:
$$i_t = \sigma\left(W_{ix}x_t + W_{ih}h_{t-1} + b_i\right)$$
- Forget gate:
$$f_t = \sigma\left(W_{fx}x_t + W_{fh}h_{t-1} + b_f\right)$$
- Candidate cell state:
$$\tilde{C}_t = \tanh\left(W_{cx}x_t + W_{ch}h_{t-1} + b_c\right)$$
- Updated cell state:
$$C_t = f_t \cdot C_{t-1} + i_t \cdot \tilde{C}_t$$
- Output gate:
$$o_t = \sigma\left(W_{ox}x_t + W_{oh}h_{t-1} + b_o\right)$$

- Hidden state:
$$h_t = o_t \cdot \tanh(C_t)$$

- Output:
$$\mathbf{y}_t = \sigma\left(W_y h_t + b_y\right)$$

4.4.1 Self Attention Long Short Term Memory

The self-attention mechanism was introduced by Vaswani et al. (2017).Once all the hidden states $\{h_1, h_2, \ldots, h_n\}$ have been computed, the self-attention mechanism is applied to identify which time steps contribute most to the final prediction. Let $H = [h_1, h_2, \ldots, h_n]^T \in R^{n\times d}$ be the matrix of all hidden states. Self-attention computes a context-aware representation:

$$\hat{H} = softmax\left(\frac{HH^T}{\sqrt{d}}\right)H$$

This weighted combination $\hat{H}$ reflects the importance of each hidden state. A global representation is obtained via pooling (e.g., average):

$$h_{final} = \frac{1}{n}\sum_{t=1}^{n}\hat{h}_t$$

The final output is then computed as:

$$y = \phi(W_y h_{final} + b_y)$$

4.4.2 Bidirectional Self Attention Long Short Term Memory

Bidirectional Recurrent Neural Networks were introduced by Schuster and Paliwal (1997). Bidirectional LSTM means that two LSTM networks are applied to the same input sequence: one processes the sequence in the forward direction (from $t = 1$ to $t = n$), using the standard LSTM equations, and the other in the backward direction (from $t = n$ to $t = 1$), using the equations below, where each time step t depends on the future time step $t + 1$.

- Input gate:
$$\overleftarrow{i_t} = \sigma\left(W'_{ix}x_t + W'_{ih}\overleftarrow{h}_{t+1} + b'_i\right)$$

- Forget gate:
$$\overleftarrow{f_t} = \sigma\left(W'_{fx}x_t + W'_{fh}\overleftarrow{h}_{t+1} + b'_f\right)$$

- Candidate cell state:
$$\overleftarrow{\tilde{C}_t} = \tanh\left(W'_{cx}x_t + W'_{ch}\overleftarrow{h}_{t+1} + b'_c\right)$$

- Updated cell state:

$$\overleftarrow{C_t} = \overleftarrow{f_t} \cdot \overleftarrow{C}_{t+1} + \overleftarrow{i_t} \cdot \overleftarrow{\tilde{C}_t}$$

- Output gate:

$$\overleftarrow{o_t} = \sigma\left(W'_{ox} x_t + W'_{oh} \overleftarrow{h}_{t+1} + b'_o\right)$$

- Hidden state:

$$\overleftarrow{h_t} = \overleftarrow{o_t} \cdot \tanh\left(\overleftarrow{C_t}\right)$$

The final hidden state at each time step is the concatenation of the forward and backward hidden states:

$$h_t = \left[\overrightarrow{h_t};\ \overleftarrow{h_t}\right]$$

The output is then computed as:

$$y_t = \phi(W_y h_t + b_y)$$

To combine the benefits of both bidirectional memory and selective focus, the BiSA-LSTM model integrates a Bidirectional LSTM with a self-attention mechanism. This architecture allows the model to capture context from both past and future time steps, and to dynamically weigh the importance of each moment in the sequence for the final prediction.

4.4.3 Neural networks use on dataset

The objective of this study is to identify the most effective neural network architecture for our dataset, with the ultimate goal of forecasting the **mean electricity prices (MA prices)**. To this end, it id evaluated several neural network models introduced previously, namely **RNN**, **LSTM**, **Self-Attention LSTM (SA-LSTM)**, and **Bidirectional SA-LSTM (BiSA-LSTM)**.

Each model is tested under various configurations, with a range of hyperparameters influencing performance. Specifically, it is varied the following:

- **Network architecture:**
  It is experimented different hidden layer configurations, including:
  - A single hidden layer with **128** neurons,
  - A single hidden layer with **256** neurons,
  - A two-layer configuration with **256** neurons in the first layer and **128** in the second.
- **Activation functions:**
  Three activation functions are evaluated:
  - **Sigmoid**,
  - **ReLU**,

- **Tanh**.

- **Timesteps:**
  It is tested how the model performs using different historical depths, by including:
  - **2**,
  - **24**,
  - **48** previous time steps as input features.

In total, over **100 unique model configurations** have been trained and evaluated.

Each configuration is assessed using **TimeSeries cross-validation**, and the models are compared based on their **Mean Absolute Error (MAE)**, defined as:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

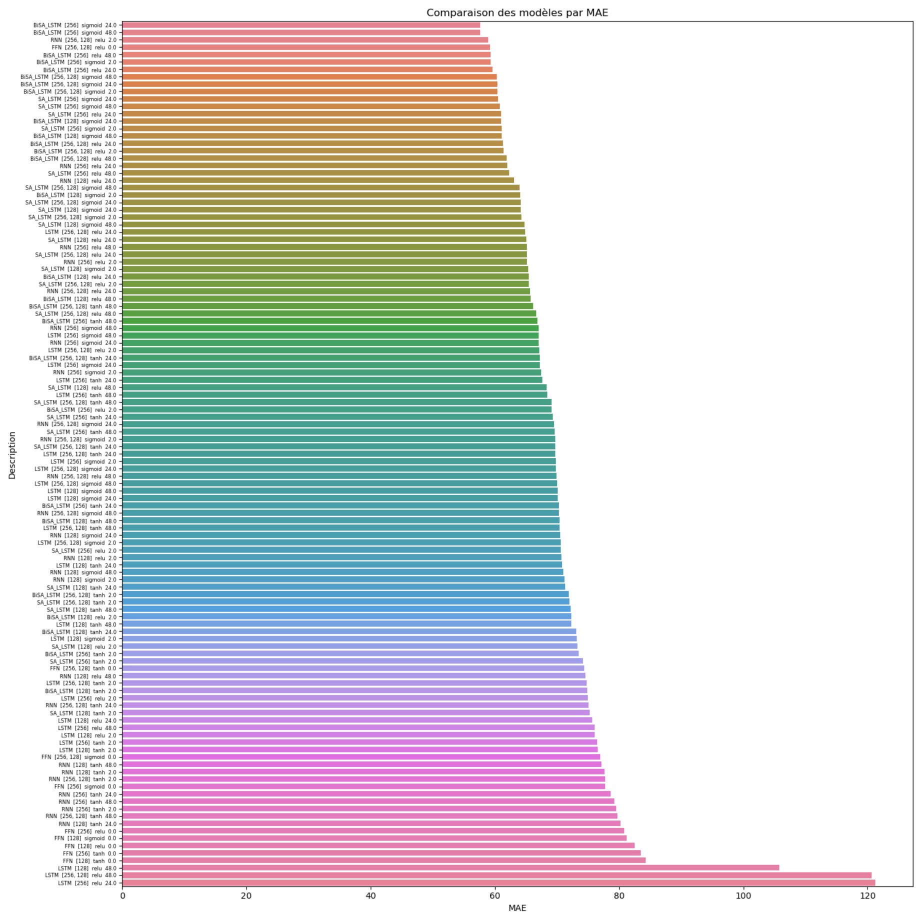

Figure 10: Model Comparison

4.4.4 Best fitted model

In the way it has been published in *Energy Economics*, Deng et al. (2024) the best-fitted model looks to be Bi-SA-LSTM model. It has the following characteristics:

- A single hidden layer with **256** neurons
- The activation function is **sigmoid**
- The timestep is **24** neurons

Now we will try to analyse this model and the results. The graph just below is a sample of the the true MA prices and the MA calculated with the model.

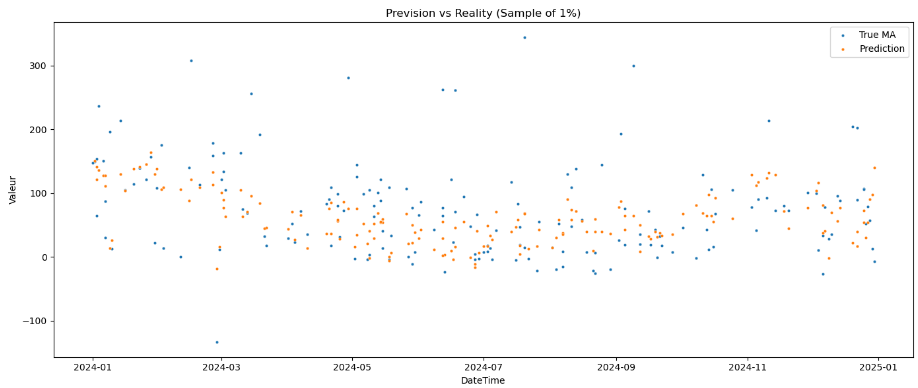

Figure 11: Bi-SA-LSTM model

On the graph it is presented only one percents of the forecasts for a better viewing. Globally the forecasts prices looks to follow the real prices. The characteristics of the forecasts are just below: Knowing that the MA price is

Modèle	**MAE**	**RMSE**	$\mathbf{R^2}$
BiSA_LSTM_256_sigmoid_24	56.4055	110.0301	0.0356

Table 2: Performance du modèle BiSA LSTM sur l'ensemble testé

centered aroud 80euros/MWh, the MAE and RMSE can be considered as high. The R2 is around 0 meaning that the model does not explain anything. It is something that can be understood by trying to evaluate the impact of each exogenous values on the MAE. Results are shown just below:

Feature	**∆ MAE**
Mois	-55.846196
WASTE	-55.867011
SOLAR	-55.878238
HYDRO_PUMPED_STORAGE	-55.887774
Heure	-55.888819
HYDRO_WATER_RESERVOIR	-55.892720
BIOMASS	-55.893699
WIND_ONSHORE	-55.894709
NUCLEAR	-55.894781
conso	-55.895031
FOSSIL_GAS	-55.895714
FOSSIL_HARD_COAL	-55.896007
HYDRO_RUN_OF_RIVER	-55.896457
SPOT	-55.897104
FOSSIL_OIL	-55.897573
Jour	-55.898266
Minute	-55.900021
Annee	-55.901871

Table 3: Feature importance on MAE

Indeed it looks like all the exogenous have the same impact on MAE. Meaning that they do not have impact on MAE. The model does not capture any real correlation. Now we will look at the evolution of the loss function during the training of the model on the dataset.

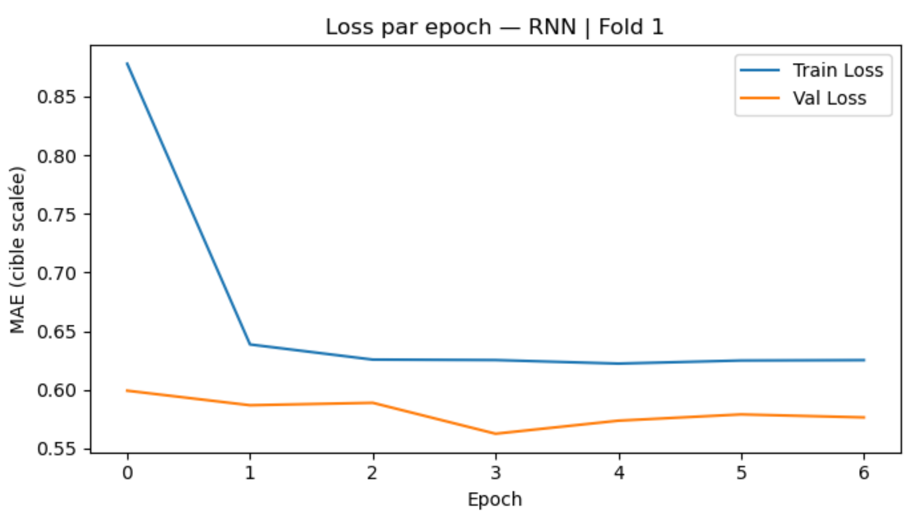

Figure 12: Loss function during the first step of the cross validation

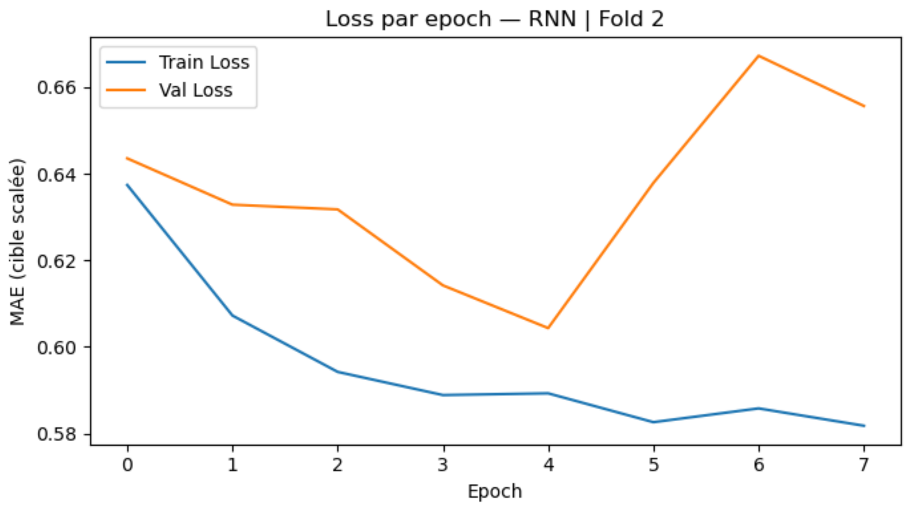

Figure 13: Loss function during the second step of the cross validation

On the figure 'Loss function during the first step of the cross validation' it is observable that train loss is decreasing during the training of the model which is what is excepted. The validation loss does not decrease so much. On the second figure the val loss is increasing fastly since the training loss is decreasing. Meaning that the model is able to reduce the errors on the dataset used for training but not on the dataset used for validation. It is like it is just learning how to obtain the best result for the data set but it is not the best way to reduce the errors on the validation set. It is a phenomenon called overfitting. It is learnt by heart but it is not useful for other dataset.
On conclusion on this part, it looks like RNN models are not well fitted for forecasting MA prices in France though they were very useful on the great britain market as mentioned in *Energy Economics*.

5 Econometrics

First of all it is going to be tested the MA prices for:

- Stationary,
- Auto-correlation,
- Homoscedasticity,
- normality.

Generally for an econometric time serie study it is used the log return. In our case and as it has been seen in the first part, the MA prices can be equal to zero and negative. So it is decided, instead of working on returns to to use the difference of MA prices. So the endogenous value that will be used is the difference MA prices.

5.1 Characteristics of the endogenous time serie

Stationary

It has been used the Dicky fuller test for testing the stationarity of the differencial prices. Results are just below:

Test Statistic	-35.4293
p-value	0.0000
Number of Lags	44
Number of Observations	17499
Critical Value (1%)	-3.4307
Critical Value (5%)	-2.8617
Critical Value (10%)	-2.5669
Additional Statistic	205717.1542

Table 4: ADF result tests

The p-value is lower than 0.05, meaning that the null hypothesis " The serie is not stationary" is rejected. So the serie is stationnary.

Serially correlated

For testing the auto correlation, it is used the Ljung Box test.

Lag	LB statistic	p-value
1	815.9628	1.8260×10^{-179}
2	1227.5049	2.8230×10^{-267}
3	1293.3716	4.0371×10^{-280}
4	1293.4359	8.8177×10^{-279}
5	1315.7705	2.4471×10^{-282}
6	1328.0229	9.2927×10^{-284}
7	1344.5954	1.3386×10^{-286}
8	1356.3429	5.7063×10^{-288}
9	1384.9864	2.4073×10^{-292}
10	1362.9654	9.6667×10^{-287}

Table 5: Ljung–Box test results for autocorrelation of differenced series

The p-value for the different lags are lower than 0.05 meaning the null hypothesis " No autocorrelation" is not respected. The series is serially correlated.

Heteroskedasticity

ARCH test is used for testing heteroskedasticity, results are just below:

Statistic	Value
LM Statistic	2695.1484
LM p-value	0.0000
F-Statistic	318.2664
F p-value	0.0000

Table 6: ARCH test results for conditional heteroskedasticity

The p-value is lower than 0.05 meaning the null hypothesis "No heteroskedasticity" is not respected.

Normality

Jacques Bera test is used for testing normality of the time serie. Results are just below:

Statistic	Value
Jarque-Bera Statistic	47379813.3096
p-value	0.0000
Skewness	-1.5558
Kurtosis	257.5690

Table 7: Jarque–Bera test results for normality

P-value is lower than 0.05, meaning the null hypothesis " The serie is normally distributed" has to be rejected.

5.2 Characteristics of the exogenous time serie

It is also studied the stationary of the exogenous variables differentiated. Results are just below:

Variable	ADF Statistic	p-value	Stationary	5% Critical Value
SPOT	-4.1199	8.9783×10^{-4}	Yes	-2.8617
BIOMASS	-6.6173	6.1596×10^{-9}	Yes	-2.8617
FOSSIL_GAS	-3.7505	3.4563×10^{-3}	Yes	-2.8617
FOSSIL_HARD_COAL	-8.7332	3.1515×10^{-14}	Yes	-2.8617
FOSSIL_OIL	-12.0830	2.2105×10^{-22}	Yes	-2.8617
HYDRO_PUMPED_STORAGE	-8.1476	9.8945×10^{-13}	Yes	-2.8617
HYDRO_RUN_OF_RIVER	-1.9977	2.8758×10^{-1}	No	-2.8617
HYDRO_WATER_RESERVOIR	-3.8281	2.6329×10^{-3}	Yes	-2.8617
NUCLEAR	-4.8007	5.4219×10^{-5}	Yes	-2.8617
SOLAR	-4.6129	1.2227×10^{-4}	Yes	-2.8617
WASTE	-4.0616	1.1201×10^{-3}	Yes	-2.8617
WIND_ONSHORE	-9.1476	2.7416×10^{-15}	Yes	-2.8617
conso	-1.8113	3.7484×10^{-1}	No	-2.8617
Prod-Conso	-2.7611	6.4045×10^{-2}	No	-2.8617

Table 8: ADF test results for stationary (critical value at 5%)

5.3 Interpretation

The different tests lead us to the following statements:

- Since the endogenous variable is stationary and serially correlated, an ARIMA model can be used to forecast the MA prices.
- The heteroskedasticity of the endogenous variable suggests using a GARCH model for forecasting volatility.
- Since the endogenous and exogenous variables are mainly stationary, a VAR model can be used.

5.4 Econometrics model

5.4.1 ARIMA Model

The ARIMA model (*AutoRegressive Integrated Moving Average*) is a classical time series forecasting approach. The ARIMA model was introduced by Box and Jenkins (1970). It combines three components:

- **Autoregressive (AR)**: regression on past values,
- **Integrated (I)**: differencing to achieve stationary,
- **Moving Average (MA)**: regression on past forecast errors.

If we consider an $ARIMA(p, d, q)$ model, after differencing the series d times to obtain y'_t, the equations to resolve are:

$$y'_t = \phi_1 y'_{t-1} + \ldots + \phi_p y'_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where:

- ϕ_i are the auto regressive coefficients,
- θ_i are the moving average coefficients,
- ε_t is the white noise error term.

5.4.2 SARIMA Model

The SARIMA model (*Seasonal ARIMA*) extends ARIMA by incorporating seasonal patterns into the model. SARIMA was introduced by HAMILTON in 1994. It is written as: The **SARIMA** (Seasonal Auto Regressive Integrated Moving Average) model is an extension of ARIMA that adds the ability to capture *seasonal patterns* in a time series. It is written as:

$$SARIMA(p, d, q) \times (P, D, Q)_s$$

where:

- (p, d, q) are the **non-seasonal** orders,
- $(P, D, Q)_s$ are the **seasonal** orders with season length s.

After applying both:

1. **Non-seasonal differencing** d times (to remove trend),
2. **Seasonal differencing** D times (to remove repeating seasonal effects),

we obtain the transformed series y''_t, and the general SARIMA equations are:

$$y''_t = \varphi_1 y''_{t-1} + \ldots + \varphi_p y''_{t-p} + \Phi_1 y''_{t-s} + \ldots + \Phi_P y''_{t-Ps} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \Theta_1 \varepsilon_{t-s} + \ldots + \Theta_Q \varepsilon_{t-Qs} + \varepsilon_t$$

where:

- φ_i are the **non-seasonal auto regressive coefficients** (short-term lags),
- Φ_i are the **seasonal auto regressive coefficients** (lags multiple of s),
- θ_i are the **non-seasonal moving average coefficients** (short-term error lags),
- Θ_i are the **seasonal moving average coefficients** (error lags multiple of s),
- ε_t is the **white noise error term**.

Key advantage: SARIMA can model both the short-term dynamics and *regularly repeating seasonal patterns* (monthly, quarterly, yearly, etc.).

5.4.3 ACF, PACF and information criteria

To find out which values of p, q, P, Q and s are the most adapted meaning which ARIMA or SARIMA model better fits the MA prices we will use Autocorrelation Function, Partial Autocorrelation Function and criteria information (AIQ, BIC and HQIC).

ACF and PACF

The autocorrelation function until 80 lags of the MA prices differentiated is shown just below.

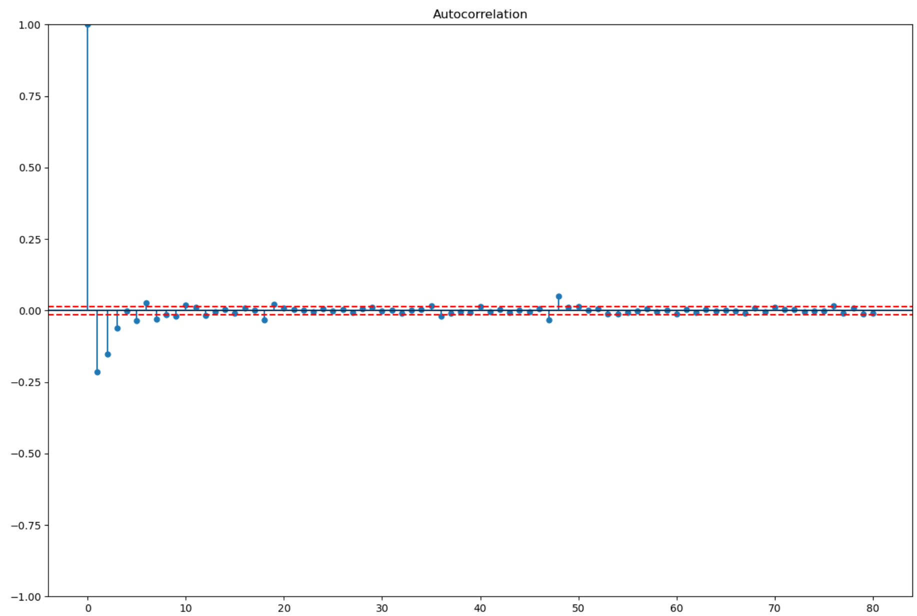

Figure 14: Autocorrelation function

Lags 1,2 and 3 are significant, giving us information on the possible order of the moving average coefficients. It also looks that lags 20 and 48 are significants. It justifys the fact to test SARIMA at order 24.

The partial autocorrelation function until 80 lags of the MA prices differentiated is shown just below.

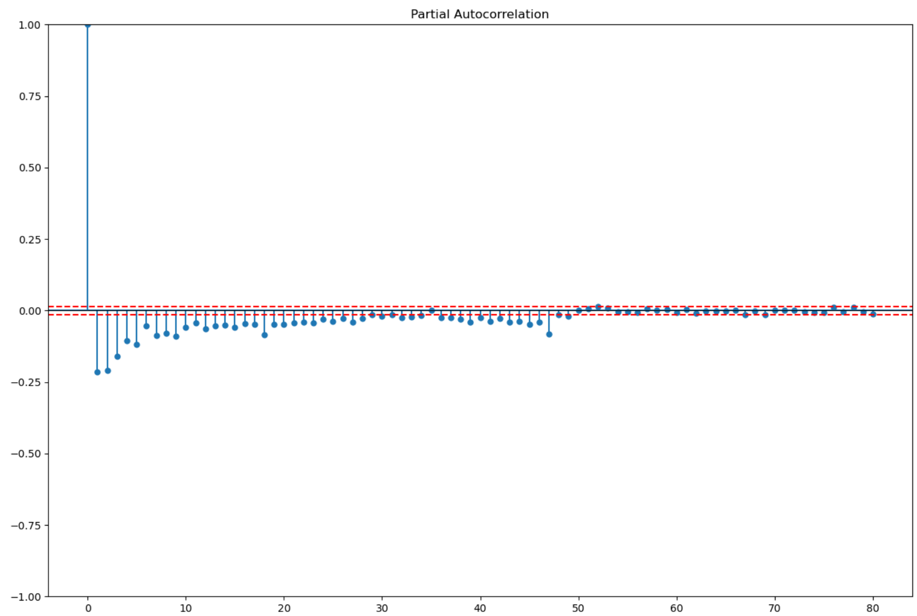

Figure 15: Partial Autocorrelation function

From lag 1 to lags 30 look significant giving information on AR order.

Information criteria

After testing orders model relevant to acf and pacf it results the table below.

Order ARIMA	AIC	BIC	HQIC
(4, 1, 2)	206262.681	206317.088	206280.595
(7, 1, 1)	206250.320	206320.272	206273.352
(4, 1, 1)	206276.839	206323.474	206292.194
(5, 1, 2)	206264.416	206326.596	206284.889
(6, 1, 2)	206257.954	206327.905	206280.986
(8, 1, 1)	206252.297	206330.021	206277.888
(3, 1, 1)	206292.590	206331.452	206305.385
(7, 1, 2)	206254.552	206332.276	206280.143
(5, 1, 1)	206278.205	206332.612	206296.119
(6, 1, 1)	206271.559	206333.739	206292.032
(8, 1, 2)	206254.162	206339.659	206282.313
(9, 1, 1)	206254.291	206339.787	206282.441
(1, 1, 1)	206325.742	206349.059	206333.419
(9, 1, 2)	206256.146	206349.415	206286.855
(2, 1, 2)	206313.162	206352.024	206325.957
(1, 1, 2)	206327.186	206358.276	206337.423
(2, 1, 1)	206327.278	206358.368	206337.515
(3, 1, 2)	206328.499	206375.133	206343.854
(9, 1, 0)	207275.377	207353.101	207300.968
(0, 1, 2)	207408.241	207431.558	207415.919
(8, 1, 0)	207419.798	207489.750	207442.830
(7, 1, 0)	207531.884	207594.063	207552.357
(6, 1, 0)	207667.248	207721.655	207685.162
(5, 1, 0)	207717.194	207763.828	207732.548
(4, 1, 0)	207968.543	208007.405	207981.339
(3, 1, 0)	208160.217	208191.307	208170.454
(2, 1, 0)	208616.259	208639.576	208623.936
(0, 1, 1)	208705.923	208721.468	208711.042
(1, 1, 0)	209400.810	209416.355	209405.929
(0, 1, 0)	210234.116	210241.889	210236.675

Table 9: Comparison information criteria for different ARIMA models

The best fitted ARIMA model is the (4,1,2) according to HQIC and BIC. It is also tested SARIMA model. Results are just below.

order	AIC	BIC	HQIC
(1, 0, 1, 20)	206250.14510	206335.64161	206278.29549
(1, 0, 1, 24)	206134.72810	206220.22461	206162.87848
(1, 0, 1, 48)	206044.70372	206130.20024	206072.85411

So finally it is going to work with the SARIMA model (7,1,1)*(1,0,1,48)

5.5 Results of the best fitted SARIMA model

Table 10: SARIMAX(7,1,1)$\times$(1,0,1)$_{48}$ results

Parameter	**Coef.**	**Std. Err.**	**z**	**P>** $\lvert z\rvert$	**[0.025**	**0.975]**
ar.L1	0.5776	0.0020	312.385	0.000	0.5740	0.5810
ar.L2	-0.0504	0.0026	-19.314	0.000	-0.0560	-0.0450
ar.L3	0.0351	0.0044	8.004	0.000	0.0270	0.0440
ar.L4	0.0359	0.0056	6.473	0.000	0.0250	0.0470
ar.L5	-0.0279	0.0043	-6.517	0.000	-0.0360	-0.0200
ar.L6	0.0514	0.0027	18.755	0.000	0.0460	0.0570
ar.L7	-0.0431	0.0037	-11.782	0.000	-0.0500	-0.0360
ma.L1	-0.9852	0.0016	-610.943	0.000	-0.9880	-0.9820
ar.S.L48	0.9695	0.0025	387.185	0.000	0.9650	0.9740
ma.S.L48	-0.9464	0.0031	-308.057	0.000	-0.9520	-0.9400
sigma^2	7804.8450	9.8180	794.924	0.000	7785.6010	7824.0890
Statistics						
Ljung-Box (L1) Q = 1.31, p-value = 0.25						
Jarque-Bera = 43 720 175.65, p-value = 0.00						
Heteroskedasticity (H) = 0.45, p-value = 0.00						
Skew = 3.33, Kurtosis = 247.86						

Regarding the table just above, several points need to be highlighted:

- All coefficients are significant since the associated p-values are lower than 0.05.
- The Ljung Box test allows us to accept the 'no autocorrelation hypothesis" of the model residuals,
- But the residuals shows heteroskedasticity

This is why a GARCH model will be added.

5.6 GARCH model

Bollerslev introduced the GARCH model in 1986. Since the homoskedasticity is not satisfied, it is necessary to model the volatility explicitly rather than assume it constant. In a GARCH model, the residuals can be expressed as:

$$\varepsilon_t = \upsilon_t \sqrt{h_t}, \quad \upsilon_t \sim \mathcal{N}(0,1)$$

with

$$h_t = \omega + \alpha_1 \varepsilon_{t-1}^2 + \ldots + \alpha_p \varepsilon_{t-p}^2 + \beta_1 h_{t-1} + \ldots + \beta_q h_{t-q}.$$

By applying a GARCH(1,1) model it is obtained:

Table 11: Results of the GARCH(1,1) model with Student-t distribution

Parameter	**Coef.**	**Std. Err.**	**t**	**P**$>\|t\|$	**[0.025**	**0.975]**
Mean Equation						
μ	-5.0159	0.2900	-17.309	4.038×10^{-67}	-5.5840	-4.4480
Volatility Equation						
ω	647.4641	47.2900	13.691	1.146×10^{-42}	554.8000	740.2000
α_1	0.5677	0.02608	21.768	4.683×10^{-105}	0.5170	0.6190
β_1	0.4323	0.01823	23.710	2.874×10^{-124}	0.3970	0.4680
Student-t Distribution						
ν	3.0414	0.06491	46.852	0.000	2.9140	3.1690
ARCH Test (Standardized Residuals)						
ARCH LM Stat = 0.0779				p-value $\approx$ 1.000		

Now the leptokurtosis and heteroskedasticity have been handled. We can now use this model and evaluate this one.

5.7 SARIMA(7,1,1)*(1,0,1,48) - GARCH(1,1)

The results of the test in sample are presented just below:

Table 12: Model performance

Metric	Value
R^2	0.99
MSE	25
MAE	5

The results leads us to the conclusion that the model performs. But for being sure of that we need to make tests out of sample. Since we want to forecast at least 24 h before it is going to use rolling windows and bootstrap to simulate real life conditions.

Out of sample

Table 13: Out-of-sample SARIMA-GARCH evaluation results

m	Wagon	R^2	MAE	MSE
100	24	-58.39	268.33	230517.67
100	48	-12.81	165.33	7794.00
100	50	-11.73	151.67	70866.33
100	75	-3.18	133.33	33236.00
100	100	-0.29	151.17	28480.19
1000	24	-8.20	117.33	23272.00
1000	48	-5.35	91.33	14107.00
1000	50	-2.28	66.67	7318.67
1000	75	-6.88	85.33	1185.00
1000	100	-8.16	2287.29	16109683.72
⋮	⋮	⋮	⋮	⋮
6000	100	-32.26	273.33	142060.67

The table above presents the out-of-sample evaluation of the SARIMA-GARCH model. Here, m denotes the starting index of the subseries extracted from the original dataset, and *Wagon* indicates the number of past observations used to train the SARIMA-GARCH model. For each experiment, the model is trained on *Wagon* points and then used to forecast the next 48 (48 lags meaning 24 hours) points of the series, corresponding to the variable *MA*. These forecasts are compared to the actual observed values using the R^2 score, the Mean Absolute Error (MAE), and the Mean Squared Error (MSE). The results show that the model performs very poorly out-of-sample, with many negative R^2 values and large error magnitudes, indicating that the SARIMA-GARCH configuration fails to capture the predictive structure of the electricity price series in this evaluation setting.

5.8 Vector Auto regressive model

Vector Auto regression (VAR).

VAR was introduced by Sims in 1980. A VAR(p) generalizes the uni variate AR model to multiple endogenous series modeled jointly. After differencing if needed to achieve stationary, a VAR(p) with exogenous regressor can be written as

$$\mathbf{y}_t \;=\; \mathbf{c} \;+\; \mathbf{A}_1\mathbf{y}_{t-1} \;+\; \cdots \;+\; \mathbf{A}_p\mathbf{y}_{t-p} \;+\; \mathbf{B}\,\mathbf{x}_t \;+\; \mathbf{u}_t, \qquad \mathbf{u}_t \sim \mathcal{N}(\mathbf{0},\, \Sigma_u),$$

where:

- $\mathbf{y}_t$ is a $k \times 1$ vector of endogenous variables (e.g., `MA`, `SPOT`, …).
- $\mathbf{A}_i$ are $k \times k$ auto regressive coefficient matrices capturing own- and cross-lag effects.
- $\mathbf{c}$ collects deterministic terms (intercept, trend, seasonal dummies).
- $\mathbf{x}_t$ is an $r \times 1$ vector of exogenous variables with coefficient matrix $\mathbf{B}$ ($k \times r$).
- $\mathbf{u}_t$ is a vector white-noise innovation with covariance Σ_u.

The lag order p is chosen via information criteria (AIC, BIC, HQIC) or lag tests. h-step-ahead forecasts are obtained recursively using the last p observed vectors and known future exogenous values:

$$\widehat{\mathbf{y}}_{t+h|t} \;=\; \mathbf{c} \;+\; \sum_{i=1}^{p} \mathbf{A}_i \, \widehat{\mathbf{y}}_{t+h-i|t} \;+\; \mathbf{B}\,\mathbf{x}_{t+h}.$$

5.8.1 Results

After differencing all the variables for being sure they are stationary. It has been applied information criteria and lag 49 has been chosen.

Here are the results:

In sample

Table 14: Significant coefficients ($p < 0.05$)

Term	Coef.	t-stat	p-val
L1.MA	0.567639	73.586	0.000
L1.SPOT	0.070284	2.182	0.029
L2.HYDRO_RUN_OF_RIVER	0.297532	2.210	0.027
L3.MA	0.035810	4.033	0.000
L3.WASTE	0.313256	1.976	0.048
L3.conso	-0.066984	-2.142	0.032
L4.MA	0.033142	3.731	0.000
L4.FOSSIL_GAS	-0.224324	-2.110	0.035
L6.MA	0.044734	5.039	0.000
L7.MA	-0.037018	-4.166	0.000
L9.FOSSIL_OIL	0.080401	3.271	0.001
L9.SOLAR	0.297718	2.407	0.016
L10.MA	0.029461	3.316	0.001
L10.SPOT	-0.104710	-2.258	0.024
L11.FOSSIL_OIL	-0.054140	-2.171	0.030
L12.FOSSIL_GAS	0.211918	1.977	0.048
L13.FOSSIL_OIL	0.053493	2.116	0.034
L17.FOSSIL_HARD_COAL	-0.271170	-2.976	0.003
L17.conso	-0.094948	-2.773	0.006
L18.MA	-0.024382	-2.754	0.006
L19.MA	0.040294	4.553	0.000
L24.FOSSIL_GAS	0.239124	2.203	0.028
L26.WASTE	0.326732	2.036	0.042
L27.FOSSIL_HARD_COAL	0.221660	2.435	0.015
L27.conso	0.085873	2.508	0.012
L28.conso	-0.072134	-2.104	0.035
L29.FOSSIL_HARD_COAL	-0.271690	-2.984	0.003
L31.FOSSIL_HARD_COAL	0.185401	2.036	0.042
L31.conso	0.157675	4.599	0.000
L32.FOSSIL_GAS	-0.215921	-1.986	0.047
L33.NUCLEAR	0.616233	4.326	0.000
L33.WIND_ONSHORE	-0.232486	-2.237	0.025
L34.BIOMASS	-0.181767	-1.974	0.048
L34.NUCLEAR	0.500734	3.516	0.000
L34.SOLAR	0.276593	2.223	0.026
L35.NUCLEAR	-0.504486	-3.543	0.000
L36.MA	-0.020247	-2.291	0.022
L36.BIOMASS	0.337411	3.673	0.000
L36.FOSSIL_GAS	-0.216727	-1.995	0.046
L36.NUCLEAR	-0.924705	-6.494	0.000
L36.WIND_ONSHORE	0.216718	2.087	0.037
L37.FOSSIL_HARD_COAL	-0.208051	-2.282	0.022
L37.NUCLEAR	0.456224	3.205	0.001
L39.BIOMASS	0.184911	2.014	0.044
L39.FOSSIL_HARD_COAL	0.229028	2.513	0.012
L40.MA	0.018051	2.043	0.041
L44.WIND_ONSHORE	0.212403	2.062	0.039
L45.FOSSIL_GAS	0.252206	2.366	0.018
L48.MA	0.060461	6.862	0.000

Table 15: In sample performance metrics

Metric	Value
R^2	0.447
MAE	41.034
MSE	6946.521

out of sample

Table 16: VAR out-of-sample performance by forecast horizon

Step	$\mathbf{R^2}$	MAE	MSE
24	-0.844	37.574	1887.026
48	-0.632	37.316	1942.714
100	-0.424	31.792	1902.645
150	-0.137	40.006	2583.999
200	-0.045	39.526	2718.898
250	-0.032	37.253	2507.629
300	0.024	39.422	2913.680
350	0.118	40.939	3499.929
400	0.190	44.139	4063.234
450	0.221	48.224	4829.717
500	-0.037	53.207	6116.511

So at the end VAR model is the one which presents the best fitted results out of sample. Though the R^2 is not so high the MAE is centered around 40 which is the MAE of the best fitted VAR model in sample. So, even if the variance is not captured by the model so well, tendency is quite well captured.

6 Machine Learning

Until now the best fitted model to our data is a VAR model which is a linear model. Now it is time to use non linear model. Famous non linear model are using decision tree. The basic idea is that at each node a split is realized into the data. A constant value is predicted on one part of the dataset and on an other part of the dataset. The point will be to minimize the MSE. Two models using these idea are Random Forest and Gradient Boosting. These models were introdced in 2001 by Breiman and Friedman. The main difference between them is that in Random Forest the tree are parallelized though in Gradient Boosting though are in added sequentially in series. Let a split be defined by variable j and threshold s. We define the two subsets:

$$R_{left}(j,s) = \{\, i \mid x_{ij} \le s \,\}, \qquad R_{right}(j,s) = \{\, i \mid x_{ij} > s \,\}.$$

The split criterion consists in choosing (j,s) that minimizes the sum of squared errors:

$$\min_{j,s} \left[\sum_{i \in R_{left}(j,s)} \bigl(y_i - \bar{y}_{left}\bigr)^2 \; + \sum_{i \in R_{right}(j,s)} \bigl(y_i - \bar{y}_{right}\bigr)^2 \right],$$

where $\bar{y}_{left}$ and $\bar{y}_{right}$ are the averages of y_i within each subset.

Random Forest

With Random Forest the predicted value is the average of the prediction of each tree. Formally, if $T_m(x)$ denotes the prediction of the m-th tree for an input x, and M is the total number of trees, the Random Forest prediction is given by:

$$\hat{y}(x) = \frac{1}{M} \sum_{m=1}^{M} T_m(x).$$

Gradient Boosting

Formally, if $h_m(x)$ is the tree fitted at iteration (getting by minimizing a loss function) m, and η is the learning rate, the model after M iterations is:

$$\hat{y}(x) = F_M(x) = F_0(x) + \sum_{m=1}^{M} \eta\, h_m(x),$$

where $F_0(x)$ is the initial prediction, typically the mean of the target variable.

6.1 Results of Random Forest and Gradient Boosting

For comparing to the VAR model it has been chosen the best fitted Random Forest model and the best fitted Gradient Boosting model. On each MAE, MSE and R^2 have been calculated in sample and out of sample. The dataset has been spited in different train and test sample. M is the proportion of the dataset which is used for testing. (1-M) is so the proportion used for training. The results are displayed just below:

Table 17: In-sample (IS) and Out-of-sample (OS) performance of Random Forest and Gradient Boosting

Model	M	R^2_{IS}	MAE_IS	MSE_IS	R^2_{OS}	MAE_OS	MSE_OS
Random Forest	0.5	0.92	15.52	1169.50	-6.29	188.39	70128.97
Random Forest	0.6	0.92	15.30	1311.47	-1.78	88.38	21551.34
Random Forest	0.7	0.92	14.51	1194.56	-0.50	75.66	11934.38
Random Forest	0.8	0.92	14.12	1122.09	-22.46	132.74	87453.51
Random Forest	0.9	0.92	13.54	1018.25	-0.45	60.62	6472.85
Gradient Boosting	0.5	0.37	55.58	9445.85	-1.56	95.65	24622.16
Gradient Boosting	0.6	0.32	55.81	10575.90	-1.39	94.95	18545.84
Gradient Boosting	0.7	0.29	54.71	10244.01	-1.47	89.20	19668.88
Gradient Boosting	0.8	0.32	54.29	9990.58	-1.22	59.55	8249.42
Gradient Boosting	0.9	0.29	52.62	9553.25	-0.38	56.72	6145.20

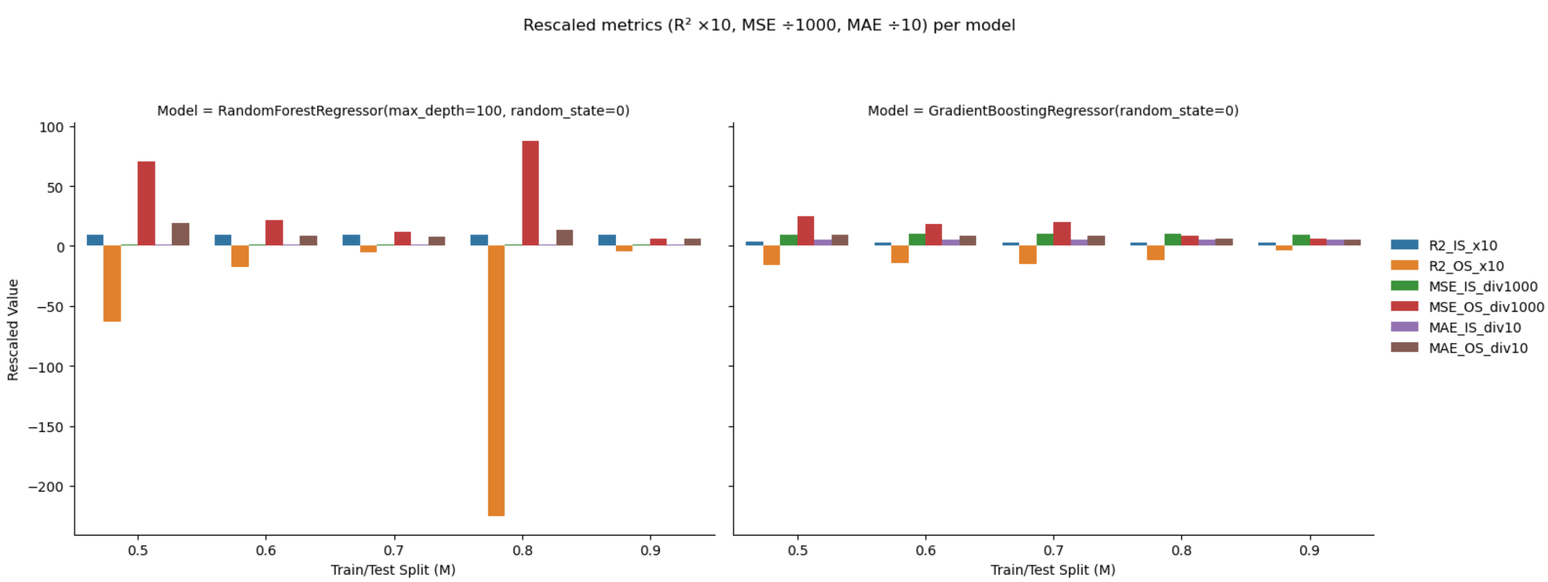

Figure 16: Random Forest and Gradient Boosting

It is displayed that Gradient Boosting is more suitable than Random Forest. The results look quite closely from the results obtained with the VAR model. Now it will be compared the VAR model and the Gradient Boosting in a real life application.

7 Can these model help to maximize the revenue of an electricity producer?

So, from now, our main conclusion is that the gradient boost and VAR models are the best-fitted models for forecasting the MA prices. But it is needed to be compared to what could be used today naively in industry. The model which will be called 'naive' is basically sum up as "the best forecast of the price at time h+24 is the price at time h".
For comparing the three models, it is going to evaluate the average absolute error for each ones. Indeed as explained previously for an energy supplier the MA prices can be used to be paid the amount such as "pay as bid". So if it is possible to better forecast the MA price, there is a possibility for the supplier to maximize revenue.
The aim is to forecast the prices at the next 24 hours, so at the next 48 lags. The MAE is calculated on several out of sample dataset. In the table below the MAE is calculated on 700 different samples. With these amount of sample the average MAE is consistent.The results are displayed just below.

	Gradient Boosting	VAR	Naive
MAE	48.5	51.3	55.2

VAR and Naive

Here are displayed the graphs of a sample taken randomly which compares the naive and the VAR model. It is observed that neither VAR or naive model perfectly fit the true values. For some cases it looks VAR model is better and for some it looks like it is Naive model which fits the best.

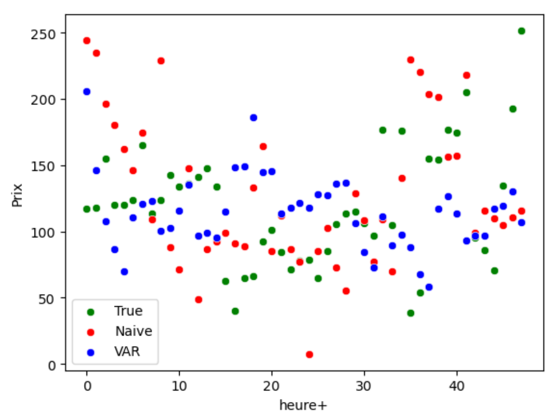

Figure 17: Price forecasting

On the graph just below it showed the difference between the foretasted and the true values. The more it is equal to zero, the better it is. It is observable that outliers are linked to the "Naive model".

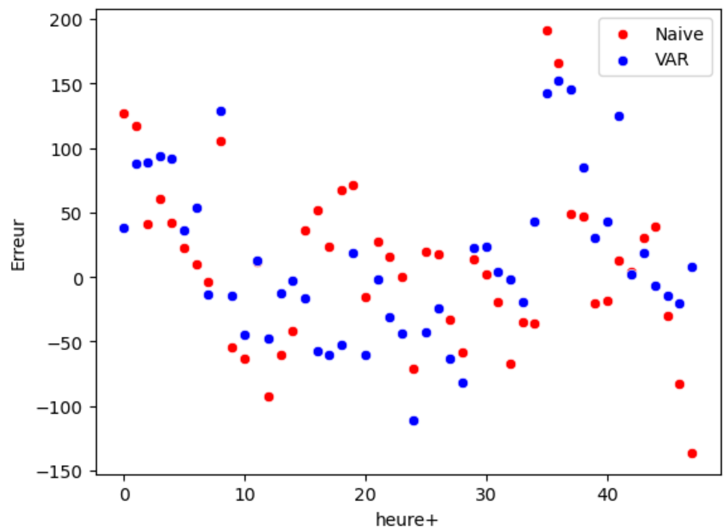

Figure 18: Forecasting gab with true value of Naive and VAR model

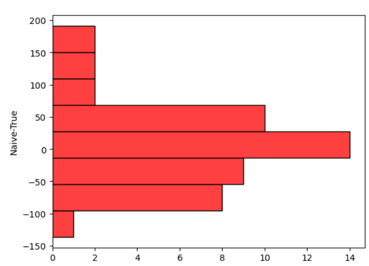

Figure 19: Naive model error histogram

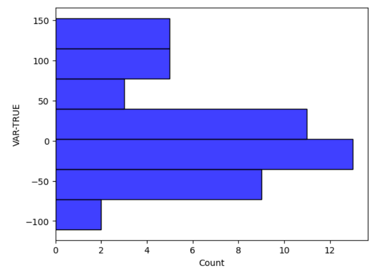

Figure 20: VAR model error histogram

Thus histograms confirms that VAR model is more efficient to minimize the gap between true and forecasted values.

Finally we can conclude that VAR is more well suited than the naive model even if the improvement is only around 8% of error reduction.

Gradient Boosting and Naive

Here are displayed the graphs of a sample taken randomly which compares the naive and the Gradient Boosting model. It is observed that neither VAR or naive model perfectly fit the true values. Globally it looks like Gradient Boosting allows to follow tendency of true values. It does not allow to fit with outliers.

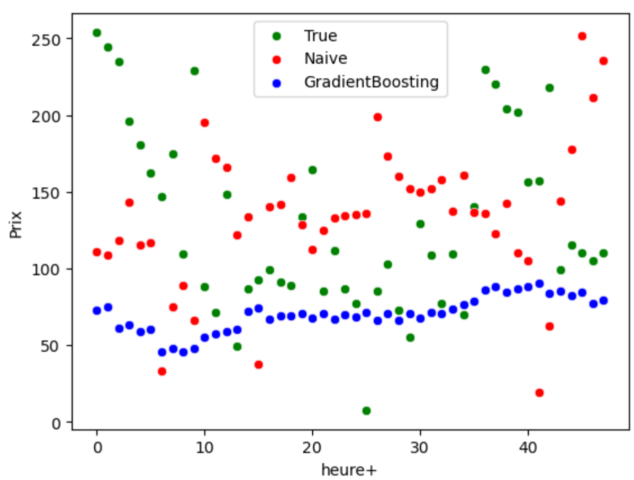

Figure 21: Gradient Boosting prices forecasting

Just below it is showed error of Gradient Boosting and Naive model. Gradient Boosting allows to largely reduce the errors.

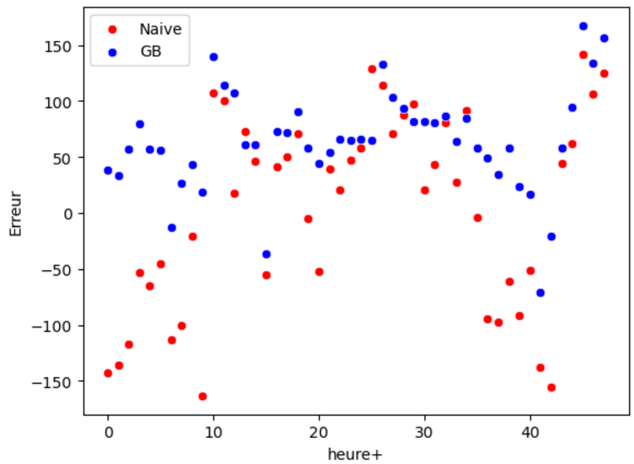

Figure 22: Gradient Boosting error forecasting

Just below it is displayed the repartition of Gradient Boosting error repartition on all the 700 samples.

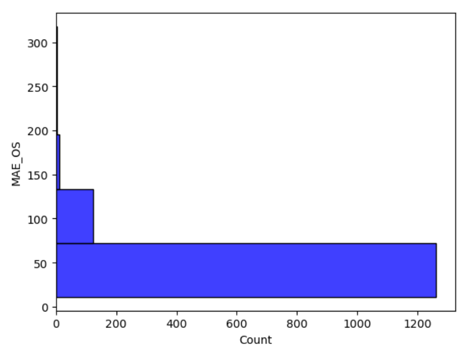

Figure 23: Gradient Boosting error repartition

Just below it is displayed the repartition of Naive model error repartition on all the 700 samples.

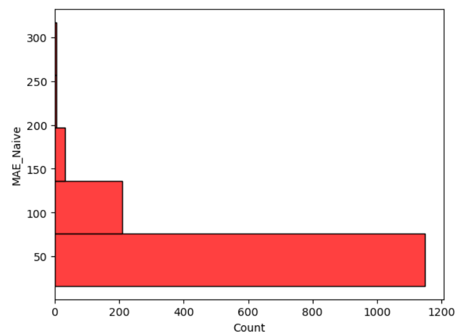

Figure 24: Naive model repartition errors

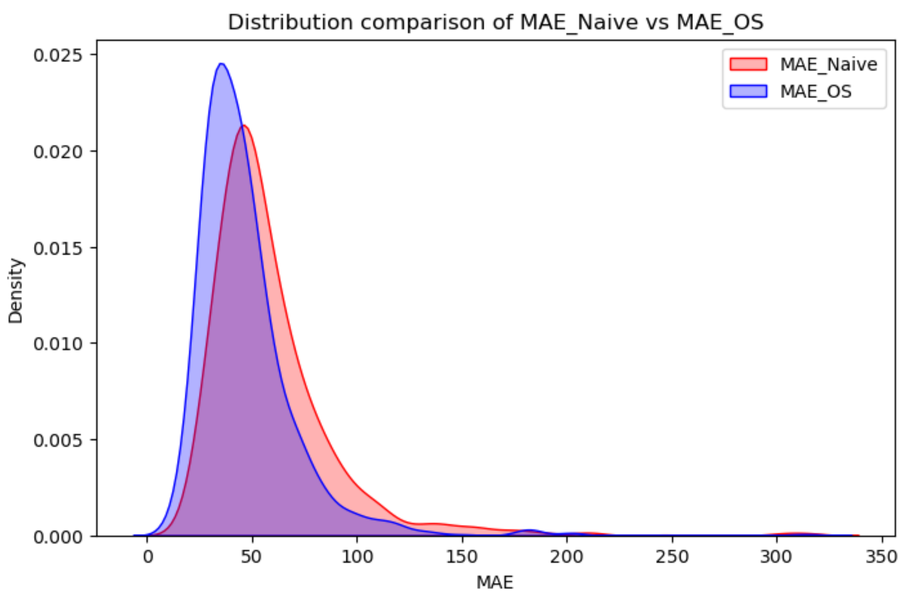

Figure 25: Comparison repartition errors

To conclude it has been showed the interest of Gradient Boosting in forecast out of sample. It is the model which minimizes the MAE. Indeed as it is observable on the figure 25, MAE is centered around 48 and the tails are smaller than ones of the naive model.

8 Conclusion

Finally, we have seen different ways for forecasting the MA prices in France. For a supplier, it is interesting to choose between selling production on the spot market or on MA. Indeed in the first one, energy is paid at the marginal price. In the second energy is payed at the bid price. So, it is definitely important for the producer to be able to forecast the MA prices for maximizing its revenue. In this paper several method have been implemented for achieving this task.

- Deep learning models,
- Econometrics model,
- Machine learning models.

For forecasting we used a dataset containing the endogenous values :

- Spot prices,
- Energy production,
- Consumption.

Though, next to the literature review it was expected that deep learning model would well fit the dataset. Nevertheless applied to our dataset the LSTM like regression were spurious. So econometrics and machine learning models have been used. Two models gave us positive forecasting out of sample:

- VAR,
- Gradient Boosting.

Forecasting out of sample using these two models have been compared to a naive model in which prices at time h+24 is equal to price at time h. A minimization of the mean absolute error around 10 % is observable.
In the end it is possible to reach a forecasting accuracy allowing to raise up the revenue of a supplier.
Model could be enriched by trying a probabilistic forecasting approach. Indeed in this paper it has only been tried to forecast values instead of intervals. Forecasting intervals could allow to increase accuracy as explained by DUTOT.

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